

# STAT 4620 Final Project Report

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## 1. Introduction

Our goal is to model the relationship between workers' wages and a set of demographic and labor-market characteristics using the Wage dataset from the ISLR package. Understanding how factors such as age, education, job classification, year, and marital status contribute to wage outcomes is important for both labor economics research and practical workforce planning. The objective is to identify a model that is both interpretable and well-supported by the data, and to summarize insights about which variables most strongly influence wages and how these effects behave across demographic groups.

## 2. Data Description and Cleaning

### 2.1 Dataset Structure

The Wage dataset includes 3000 observations with 13 variables. Table 1 shows the description and measurement type for each variable.

Table 1: Variable Descriptions for Wage Dataset

| Variable   | Description                                                                                             | Type                     |
|------------|---------------------------------------------------------------------------------------------------------|--------------------------|
| X          | Respondent ID (unique identifier)                                                                       | Numerical:<br>Discrete   |
| year       | Year that wage information was recorded                                                                 | Numerical:<br>Discrete   |
| age        | Age of worker                                                                                           | Numerical:<br>Discrete   |
| maritl     | Marital status: 1. Never Married, 2. Married, 3. Widowed, 4. Divorced, 5. Separated                     | Categorical:<br>Nominal  |
| race       | Race of worker: 1. White, 2. Black, 3. Asian, 4. Other                                                  | Categorical:<br>Nominal  |
| education  | Highest education level: 1. < HS Grad, 2. HS Grad, 3. Some College, 4. College Grad, 5. Advanced Degree | Categorical:<br>Ordinal  |
| region     | Region of country (mid-atlantic only)                                                                   | Categorical:<br>Nominal  |
| jobclass   | Job class: 1. Industrial, 2. Information                                                                | Categorical:<br>Nominal  |
| health     | Self-reported health status: 1. <=Good, 2. >=Very Good                                                  | Categorical:<br>Ordinal  |
| health_ins | Health insurance status: 1. Yes, 2. No                                                                  | Categorical:<br>Binary   |
| logwage    | Log-transformed workers wage                                                                            | Numerical:<br>Continuous |
| wage       | Workers raw wage                                                                                        | Numerical:<br>Continuous |
| Resp       | Mystery response variable                                                                               | Numerical:<br>Continuous |

### 2.2 Data Cleaning & Preprocessing

All original variables are complete. The appended variable Resp contains missing values. To examine potential bias, we created an indicator variable marking whether Resp was missing. Comparing summary

statistics of numerical predictors between missing and non-missing Resp cases showed similar mean values, suggesting that the missingness is likely missing at random (Table 2). Therefore, removing rows with missing Resp will not bias the analysis. We also checked for duplicate rows and found none. Thus, our final cleaned dataset excludes the identifier variable X and all rows with missing Resp.

Table 2: Mean of Numeric Variables by Missingness of Resp

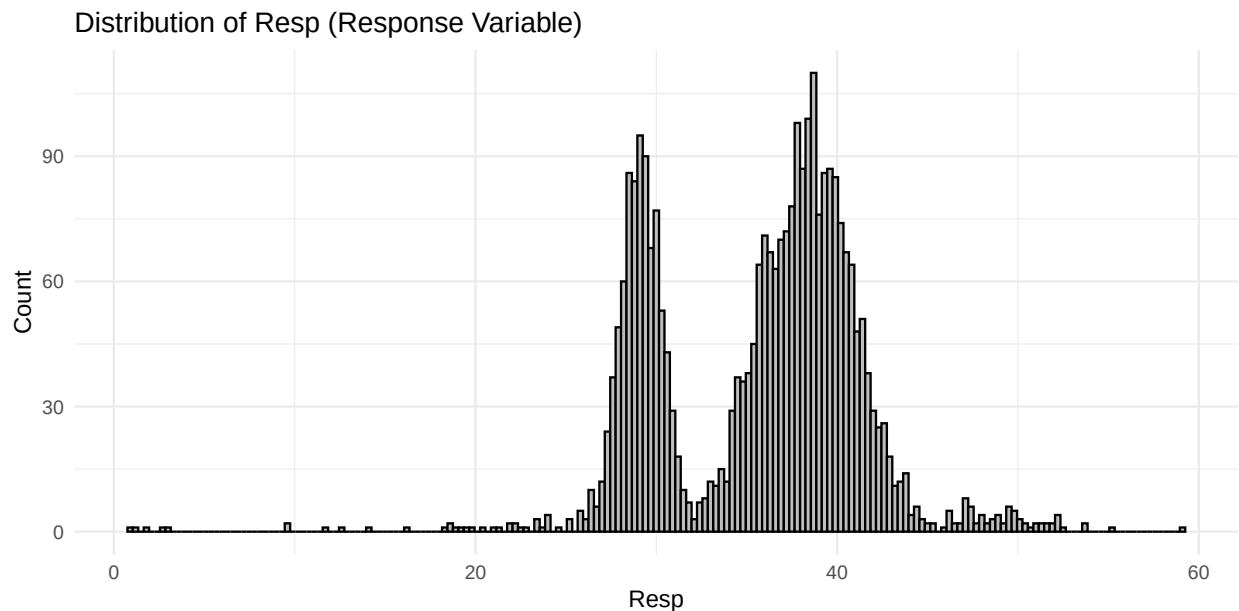
| missing | X        | year    | age   | logwage | wage   | Resp  |
|---------|----------|---------|-------|---------|--------|-------|
| FALSE   | 219006.4 | 2005.79 | 42.42 | 4.65    | 111.73 | 35.66 |
| TRUE    | 212854.1 | 2005.62 | 42.15 | 4.67    | 110.65 | NaN   |

```
data_clean <- data %>%
  dplyr::select(-X) %>%
  dplyr::filter(!is.na(Resp))
```

### 3. Exploratory Data Analysis

#### 3.1 Distribution of the Response Variable (Resp)

Before examining predictors, we examine the shape of the response variable. The histogram below shows that **Resp** is clearly **bimodal**, with two distinct peaks.



#### Interpretation

The strong **bimodal distribution** suggests that:

- A linear mean structure may not fully capture the data.
- Some predictors may shift individuals into one mode or the other.

This motivates the need for polynomial and semi-parametric models.

## 3.2 Summary of Numeric Variables & Categorical Variables

The variable wage shows a wide range, with a median that is lower than the mean, indicating the presence of some high earners. The variable logwage is the logarithmic transformation of wage, which reduces skewness and makes the distribution more symmetric (Table 3). The categorical variables were summarized by their most frequent categories (Table 4).

Table 3: Summary Statistics – Numeric Variables

| Variable | Min     | Median  | Mean    | Max     |
|----------|---------|---------|---------|---------|
| year     | 2003.00 | 2006.00 | 2005.79 | 2009.00 |
| age      | 18.00   | 42.00   | 42.42   | 80.00   |
| logwage  | 3.00    | 4.65    | 4.65    | 5.76    |
| wage     | 20.09   | 104.92  | 111.73  | 318.34  |
| Resp     | 1.00    | 36.95   | 35.66   | 59.11   |

Table 4: Most Frequent Category – Categorical Variables

| Variable   | Most Frequent Category |
|------------|------------------------|
| maritl     | 2. Married             |
| race       | 1. White               |
| education  | 2. HS Grad             |
| region     | 2. Middle Atlantic     |
| jobclass   | 1. Industrial          |
| health     | 2. >=Very Good         |
| health_ins | 1. Yes                 |

## 3.3 Correlation Analysis

The variables wage and logwage are highly correlated, as expected due to the log transformation. The response variable Resp shows moderate positive correlations with wage, logwage, and age, suggesting that these variables may be influential in predicting Resp. The year variable shows a weak correlation with Resp, indicating limited time-based effects.



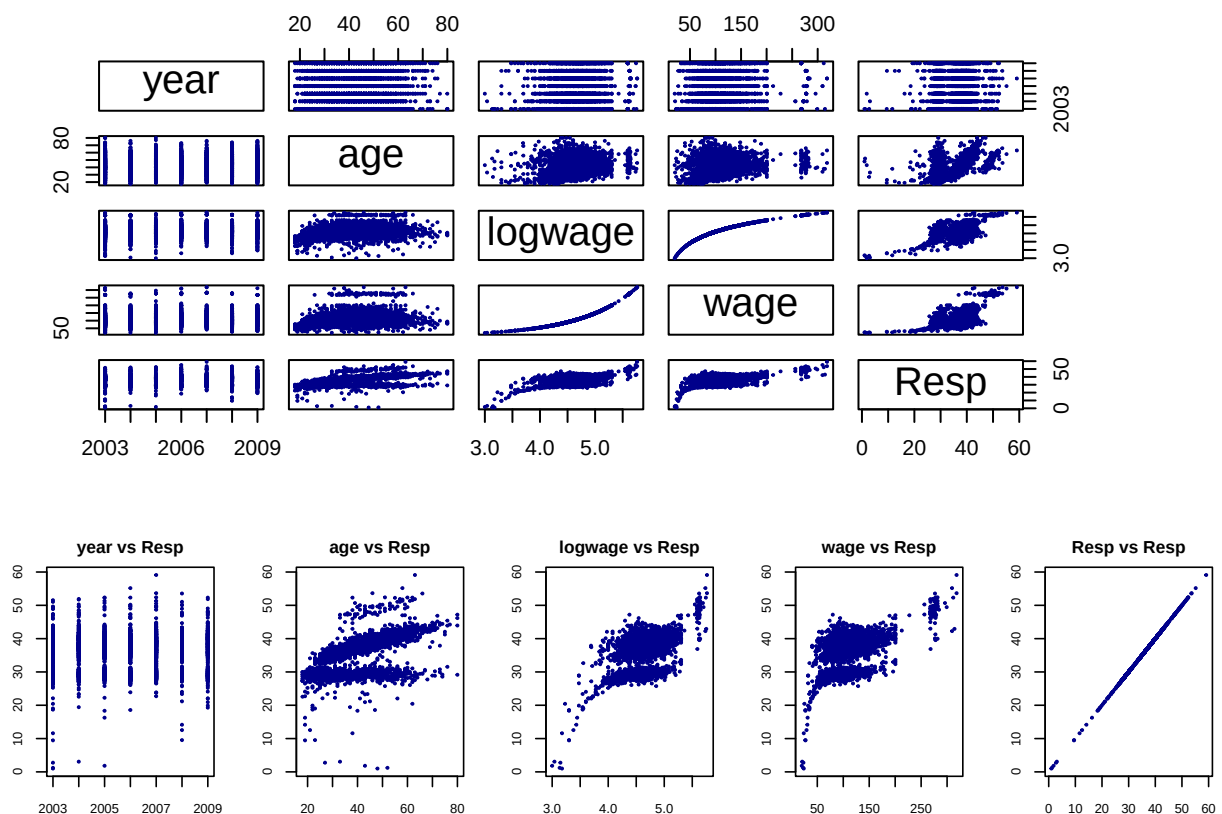
## 3.4 Numeric Variables vs Response

From scatterplot matrix, there is a positive, non-linear relationship. Wages tend to increase with age up to around 60, after which they seem to level off or slightly decrease. The variance also appears to increase with

age. The plot also confirms that Resp is highly correlated with logwage and wage, and positively related to age.

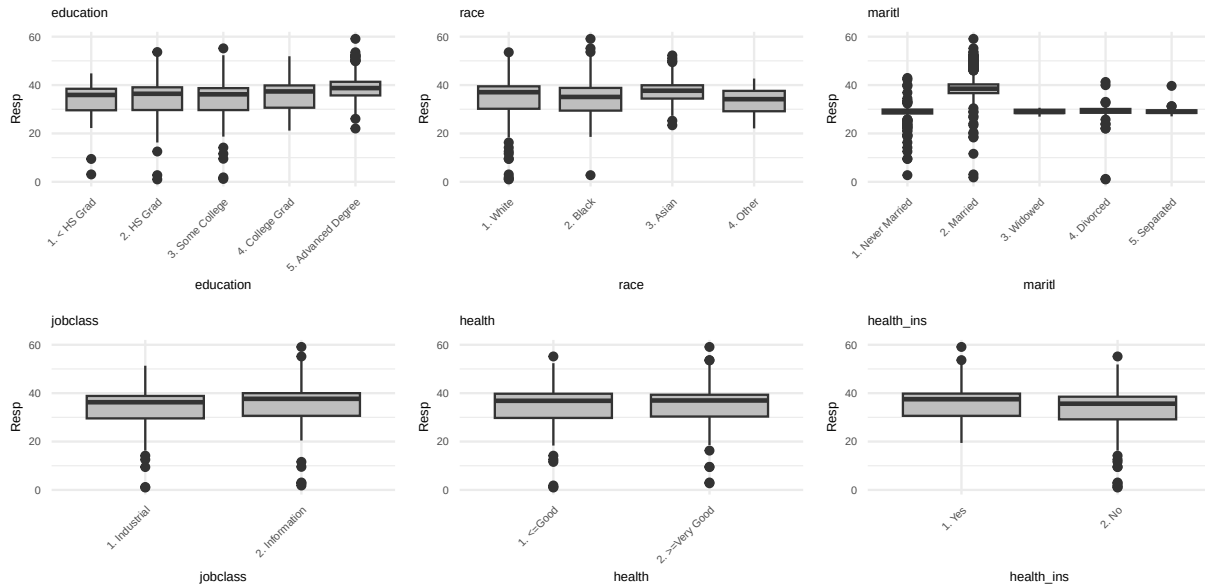
When we specifically look at the relationship between Resp and the continuous predictors:

- age vs. Resp: The three apparent horizontal lines or clusters are most likely caused by the presence of a strong categorical predictor. A small number of highly influential categories in a predictor variable can create distinct, separated bands in a scatter plot when plotted against any continuous variable (like age).
- logwage/Wage vs. Resp: The plots are separated into distinct clusters primarily due to the influence of a strong categorical predictor.



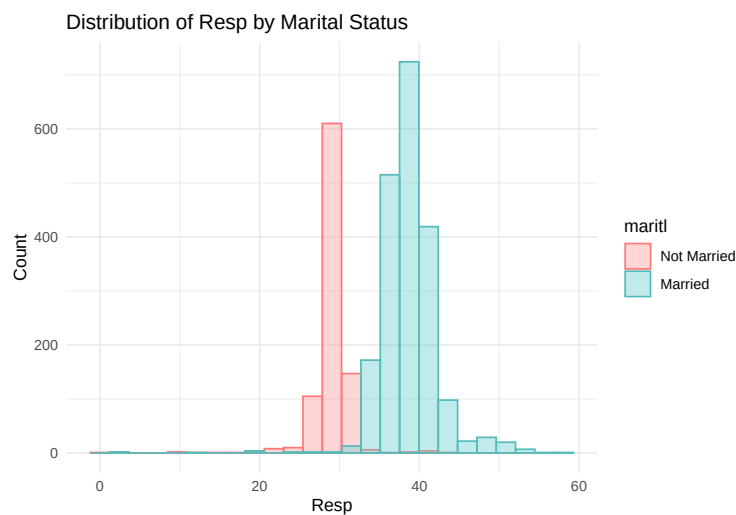
### 3.5 Categorical Variables vs Response

- education: Clear, strong separation between levels. Higher education corresponds to higher Resp.
- maritl (Married vs Unmarried): This is one of the strongest relationships in the data. Married individuals disproportionately fall into the upper mode of the Resp distribution. This key observation may explain the bimodal distribution and motivates the strong reliance on the marital interaction term in the subsequent modeling sections.
- race, health, health\_ins: These show weaker group separation. Their effects appear smaller and do not meaningfully explain the bimodality of Resp.
- jobclass: Practically no separation in Resp; likely not an important predictor.
- Outlier Identification: The outliers show data points with unusually low or high wages for their respective categories, which may warrant further investigation for potential influential observations.

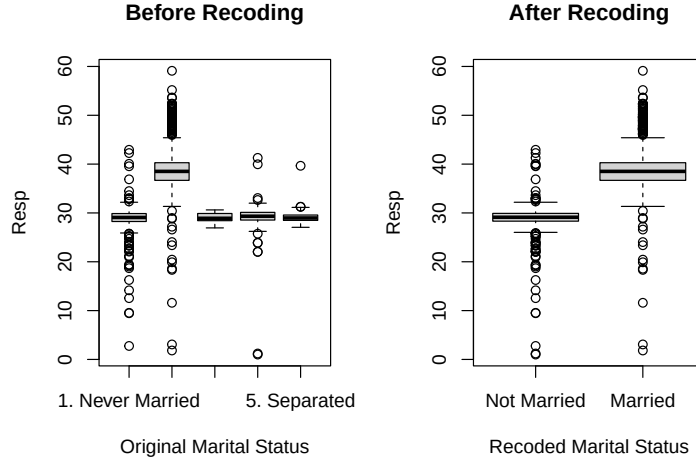


### 3.6 Recoding Marital Status (Married vs. Unmarried)

The boxplots on section 3.5 show a clear distinction between the married and unmarried classes of the marital variable in relationship to the response variable. After coding the marital variable as binary (1 = Married, 0 = Unmarried), plotting the response and coding it according to the new binary variable, we see that the two distinct modes in the bimodal response appear to be explained by marital status.



Therefore, for the remainder of this report, marital will be modeled as a binary variable.



### 3.7 Variables interactions with Age and Logwage

Our earlier results suggested that both age and logwage are strong predictors of the response variable, so we examined whether their effects vary across different subgroups.

#### Interactions with Age

The interaction tests show clear evidence that the relationship between **age** and the response varies across several demographic and health-related variables. The strongest effect appears in the **age**  $\times$  **maritl** interaction, followed by meaningful interactions with **health** and **health\_ins**. These findings indicate that the effect of age is not uniform; instead, it depends on an individual's marital status, health condition, and insurance coverage—especially marital status. (Appendix 1)

Significant interactions with age:

- age  $\times$  **maritl** (strongest)
- age  $\times$  **health**
- age  $\times$  **health\_ins**

The interaction between age  $\times$  **maritl** is highly significant ( $p < 0.001$ ), confirming that marital status meaningfully alters the age effect on the response. Interactions with **health** and **health\_ins** are also statistically significant, though to a lesser degree, indicating that age's effect varies in modest but important ways across these groups. (Appendix 3)

Table 5: ANOVA Interaction Summary

| Interaction                    | F      | p-value     | Interpretation      |
|--------------------------------|--------|-------------|---------------------|
| age $\times$ <b>maritl</b>     | 69.738 | $< 2.2e-16$ | Highly significant. |
| age $\times$ <b>health</b>     | 8.324  | 0.00394     | Significant.        |
| age $\times$ <b>health_ins</b> | 5.570  | 0.0183      | Significant.        |

#### Interactions with Logwage

The interaction analysis for **logwage** shows a different pattern. The strongest and most meaningful interaction is with **health\_ins**, indicating that the relationship between wages and the response variable differs considerably between individuals with and without health insurance. Marital status shows a marginally significant interaction, while education, jobclass, and health do not appear to moderate the logwage–response relationship. (Appendix 2)

Significant interactions with logwage:

- $\text{logwage} \times \text{health\_ins}$  (strongest)
- $\text{logwage} \times \text{maritl}$  (marginal)

For logwage, the interaction with health\_ins again emerges as highly significant ( $p < 0.001$ ), showing that the wage effect differs sharply depending on health insurance status. The  $\text{logwage} \times \text{maritl}$  interaction is marginally significant ( $p = 0.047$ ), suggesting that while there may be small differences across marital groups, the effect is weaker. (Appendix 4)

Table 6: ANOVA Interaction Summary

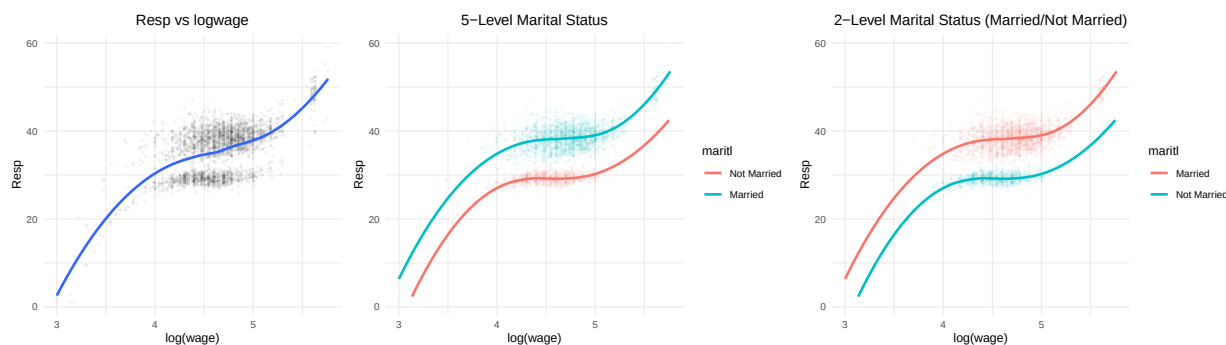
| Interaction                                | F      | p-value   | Interpretation                                |
|--------------------------------------------|--------|-----------|-----------------------------------------------|
| $\text{logwage} \times \text{health\_ins}$ | 18.858 | 1.456e-05 | Interaction significantly improves the model. |
| $\text{logwage} \times \text{maritl}$      | 2.416  | 0.04675   | Interaction barely significant.               |

Our exploratory interaction tests confirmed that the effect of continuous predictors is not uniform. The most critical interaction, which became the cornerstone of our best models, is the one involving marital status.

### 3.8 Understanding Interactions

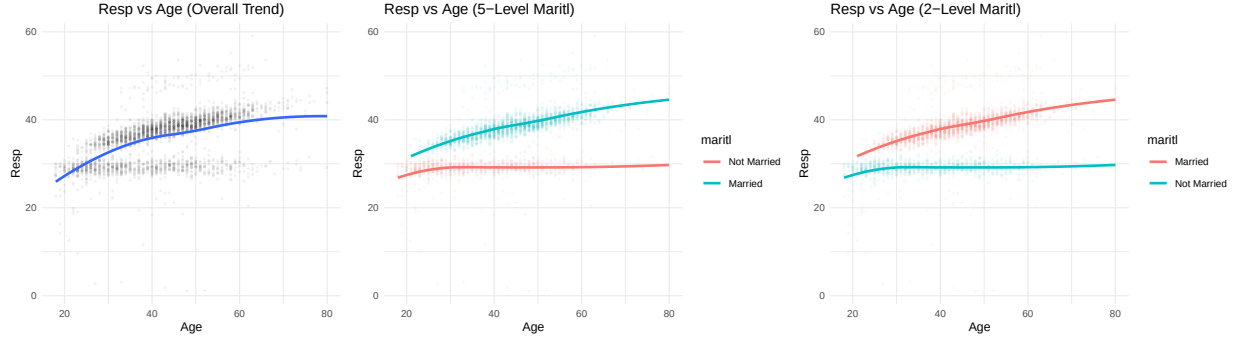
#### Resp vs $\log(\text{Wage})$

While the model without marital status shows one smooth upward trend, the inclusion of marital status produces two distinct trajectories, especially at higher wage levels, illustrating a meaningful interaction effect.



#### Resp vs Age

Including marital status reveals an interaction effect, where married individuals display consistently higher and more rapidly increasing response values across age.



## 4. Modeling Strategy and Results

### 4.1 Data Partitioning and Modeling Approach

The cleaned data ( $n = 2,940$ ) was partitioned into an **80% training set** (Wage.train) for model fitting and hyperparameter tuning, and a **20% test set** (Wage.test) for unbiased performance evaluation. Our modeling strategy moved from simple parametric models to complex non-parametric methods to find the optimal balance between predictive accuracy (low Test MSE) and statistical inference (interpretability):

- **Baseline & Regularized Models:** (MLR, Ridge, LASSO) provide a simple linear benchmark.
- **Interpretable Non-Linear Models:** (Polynomial Regression, GAMs) capture non-linear trends and key interactions.
- **Non-Parametric Ensemble Models:** (Random Forest, Boosting) offer maximum predictive power by capturing arbitrary interactions.

### 4.2 Performance of Interpretable Non-Linear Models

#### 4.2.1 Polynomial Regression:

Polynomial regression was first used as a simple and interpretable way to introduce curvature in the model.

- Captures curvature without fully abandoning linear structure
- Used as a first benchmark for nonlinearity
- Fit with and without marital-status interaction terms
- Helps determine whether interaction effects meaningfully improve model accuracy

#### Polynomial Regression Results

The Best Polynomial Model used a third-degree polynomial for logwage and a second-degree polynomial for age, with both terms fully interacting with maritl (recoded as Married/Not Married), plus all other categorical predictors.

$$\text{Resp} \sim \text{poly}(\text{logwage}, 3) \times \text{maritl} + \text{poly}(\text{age}, 2) \times \text{maritl} + \text{All Categoricals}$$

Table 7: log(Wage) and Age Polynomial Test MSE Results

| Model | Test_MSE |
|-------|----------|
|-------|----------|

|                                                           |           |
|-----------------------------------------------------------|-----------|
| Age Polynomial (Degree 2)                                 | 20.910368 |
| log(Wage) Polynomial (Degree 3)                           | 19.190456 |
| Age Polynomial (Degree 2) with Marriage Interaction       | 6.202241  |
| log(Wage) Polynomial (Degree 3) with Marriage Interaction | 4.079964  |
| Age and log(wage) Polynomial (no interaction)             | 13.610247 |
| Age and log(wage) with marital status interaction         | 1.035718  |

The results show that adding nonlinear terms for both predictors does not improve performance by much unless interaction effects are included, indicating that subgroup differences by marital status drive much of the predictive gain.

### Introducing Categorical Predictors

In an effort to improve the model’s performance, the age and log(Wage) model with interaction effects is set as the baseline. From there, various categorical values were tested to gauge an improvement in Test MSE.

Table 8: Top 5 Polynomial Models by Test MSE

| Model                                           | Test_MSE |
|-------------------------------------------------|----------|
| Base + race + jobclass                          | 0.979    |
| Base + race + jobclass + health_ins             | 0.979    |
| Base + education + race + jobclass              | 0.980    |
| Base + education + race + jobclass + health_ins | 0.980    |
| Base + jobclass                                 | 0.980    |

The top-performing polynomial models all include the full or near-full set of categorical predictors, suggesting that demographic and job-related characteristics contribute additional explanatory value beyond nonlinear age and wage effects.

### 4.2.2 Generalized Additive Models (GAM):

GAMs were implemented to address the limitations of polynomial models by learning smooth, data-driven functions rather than relying on fixed polynomial shapes.

- Allows for smooth functions rather than assuming polynomial shape
- Supports different smooths by marital status, aligning with observed interaction effects
- Provides interpretability while modeling nonlinear patterns more accurately
- Iteratively expanded to include additional categorical predictors (education, race, jobclass)

### Generalized Additive Models (GAM) Results:

A Generalized Additive Model (GAM) was then fit to replicate the baseline structure used in the polynomial models, incorporating log(Wage) and Age along with their interactions with marital status. Unlike polynomial regression, GAM allows the use of smoothing spline functions, enabling a flexible, non-parametric fit that can better capture nonlinear patterns without imposing a fixed global shape.

The Best GAM used smoothing splines (s) for logwage and age, with their smooth functions varying based on marital status (known as gam12 in the internal analysis):

$$\text{Resp} \sim s(\text{logwage}, \text{by} = \text{maritl}) + s(\text{age}, \text{by} = \text{maritl}) + \text{All Categoricals}$$

Table 9: Top 5 GAM Models by Test MSE

| Model                                           | Test_MSE |
|-------------------------------------------------|----------|
| Base + race + jobclass + health_ins             | 0.982    |
| Base + race + jobclass                          | 0.982    |
| Base + education + race + jobclass              | 0.983    |
| Base + education + race + jobclass + health_ins | 0.983    |
| Base + jobclass                                 | 0.984    |

Similar to polynomial regression, these results suggest that the GAM framework benefits from a full or nearly full list of additional categorical predictors, which enhance predictive accuracy when paired with spline-based nonlinear effects.

### Comparison to Polynomial Regression

Compared to the polynomial regression models, the GAM produced slightly higher Test MSE, likely because its smoother spline-based functions constrain flexibility and introduce bias by not attempting to capture sharp curvature in the data. This slight increase in bias does result in a slightly lower residual variance, as observed in the table below.

Table 10: Comparison of Best Polynomial and GAM Models

| Metric             | Best.Polynomial | Best.GAM |
|--------------------|-----------------|----------|
| Test MSE           | 0.982           | 0.985    |
| 10-fold CV MSE     | 1.028           | 1.041    |
| Residual Variance  | 1.010           | 1.010    |
| Total Effective DF | 22.000          | 31.480   |

The best polynomial model achieves slightly lower test MSE (0.976 vs 0.986) and lower 10-fold CV MSE (1.031 vs 1.056) than the GAM, indicating marginally better out-of-sample performance and stability with fewer effective degrees of freedom (22 vs 31.11). Although the GAM attains a very slightly lower residual variance, this comes at the cost of greater complexity, so the polynomial model offers a better balance of accuracy, parsimony, and interpretability.

### Tree Based Method Results

Description as to why...

#### Include Table Here for 4 Methods

- Test MSE
- CV MSE
- Residual Variance
- Effective Degrees of Freedom

Short summary of best one.

### Comparison to Parametric Models

Explain pros but mostly cons because it doesn't perform as well

## Model Evaluation

- Explain how close the polynomial and GAM are in results (compare bias and variance). Lowkey this is basically the section I did above (Summary Comparison of Best Polynomial and GAM Models) so maybe it would be better here – perhaps a copy paste scenario. You’d have to add the tree results too, just pick the best of the tree methods (boosting I think)
- Looks like Polynomial has the edge in Test MSE and CV MSE. Although it barely loses the variance battle, its effective DF is so much lower than the GAM spline method.
- Talk about explainability of polynomial method vs others.
- Write out final Model Equation

## Appendix

### Appendix 1

```
##
## Call:
## lm(formula = Resp ~ age * education, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -36.464  -2.133   1.460   2.786  18.285
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    24.120395    1.006477   23.965 < 2e-16 ***
## age             0.247916    0.023062   10.750 < 2e-16 ***
## education2. HS Grad    0.967445    1.152390    0.840  0.40125
## education3. Some College -0.122795    1.220315   -0.101  0.91985
## education4. College Grad  1.348483    1.243922    1.084  0.27843
## education5. Advanced Degree  4.274717    1.445696    2.957  0.00313 **
## age:education2. HS Grad  -0.012724    0.026357   -0.483  0.62930
## age:education3. Some College  0.015126    0.028214    0.536  0.59190
## age:education4. College Grad  0.002141    0.028392    0.075  0.93989
## age:education5. Advanced Degree -0.025325    0.032253   -0.785  0.43241
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.714 on 2930 degrees of freedom
## Multiple R-squared:  0.2984, Adjusted R-squared:  0.2962
## F-statistic: 138.5 on 9 and 2930 DF, p-value: < 2.2e-16

##
## Call:
## lm(formula = Resp ~ age * jobclass, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -36.410  -2.284   1.420   2.679  17.964
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    24.41599    0.45339   53.852 <2e-16 ***
## age             0.25397    0.01054   24.097 <2e-16 ***
## jobclass2. Information  1.50252    0.67798    2.216  0.0268 *
## age:jobclass2. Information -0.01231    0.01539   -0.800  0.4238
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```

## Residual standard error: 4.786 on 2936 degrees of freedom
## Multiple R-squared:  0.2756, Adjusted R-squared:  0.2748
## F-statistic: 372.3 on 3 and 2936 DF,  p-value: < 2.2e-16

##
## Call:
## lm(formula = Resp ~ age * health, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -35.719  -2.321   1.305   2.755  18.574
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    25.06366    0.62412  40.159 < 2e-16 ***
## age            0.22823    0.01338  17.051 < 2e-16 ***
## health2. >=Very Good   -0.66872    0.74268  -0.900  0.36798
## age:health2. >=Very Good  0.04718    0.01635   2.885  0.00394 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.763 on 2936 degrees of freedom
## Multiple R-squared:  0.2824, Adjusted R-squared:  0.2816
## F-statistic: 385 on 3 and 2936 DF,  p-value: < 2.2e-16

##
## Call:
## lm(formula = Resp ~ age * maritl, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -36.432  -0.963  -0.092   0.816  16.549
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    27.877220    0.296973  93.871 < 2e-16 ***
## age            0.028096    0.007630   3.682 0.000235 ***
## maritlMarried  1.153880    0.406650   2.838 0.004578 **
## age:maritlMarried 0.186620    0.009729  19.183 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.809 on 2936 degrees of freedom
## Multiple R-squared:  0.7504, Adjusted R-squared:  0.7501
## F-statistic: 2942 on 3 and 2936 DF,  p-value: < 2.2e-16

##
## Call:
## lm(formula = Resp ~ age * health_ins, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -36.222  -2.315   1.338   2.704  18.296
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    26.246081    0.433896  60.489 < 2e-16 ***
## age            0.231193    0.009672  23.903 < 2e-16 ***
## health_ins2. No  -2.799121    0.687987  -4.069 4.85e-05 ***
## age:health_ins2. No  0.037786    0.016010   2.360  0.0183 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.773 on 2936 degrees of freedom
## Multiple R-squared:  0.2794, Adjusted R-squared:  0.2786
## F-statistic: 379.4 on 3 and 2936 DF,  p-value: < 2.2e-16

```

## Appendix 2

```
##
## Call:
## lm(formula = Resp ~ logwage * education, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -21.4318  -3.4202   0.4979   3.3329  14.5442
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -7.5057     4.5991  -1.632   0.1028
## logwage         9.5466     1.0439   9.146 <2e-16 ***
## education2. HS Grad  0.1344     5.1380   0.026   0.9791
## education3. Some College -11.6923     5.3420  -2.189   0.0287 *
## education4. College Grad  5.1910     5.2524   0.988   0.3231
## education5. Advanced Degree -6.7308     5.6107  -1.200   0.2304
## logwage:education2. HS Grad -0.1630     1.1599  -0.141   0.8883
## logwage:education3. Some College  2.0911     1.1966   1.748   0.0807 .
## logwage:education4. College Grad -1.4812     1.1709  -1.265   0.2060
## logwage:education5. Advanced Degree  1.0683     1.2280   0.870   0.3844
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.556 on 2930 degrees of freedom
## Multiple R-squared:  0.3448, Adjusted R-squared:  0.3428
## F-statistic: 171.4 on 9 and 2930 DF, p-value: < 2.2e-16

##
## Call:
## lm(formula = Resp ~ logwage * jobclass, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -21.5334  -3.4330   0.4929   3.4276  13.9825
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -5.4762     1.6410  -3.337 0.000857 ***
## logwage         8.8189     0.3572  24.689 < 2e-16 ***
## jobclass2. Information -1.6739     2.3052  -0.726 0.467812
## logwage:jobclass2. Information  0.3954     0.4941   0.800 0.423594
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.618 on 2936 degrees of freedom
## Multiple R-squared:  0.3253, Adjusted R-squared:  0.3246
## F-statistic: 471.8 on 3 and 2936 DF, p-value: < 2.2e-16

##
## Call:
## lm(formula = Resp ~ logwage * health, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -21.2496  -3.3901   0.4582   3.3853  13.6121
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -7.6387     2.1353  -3.577 0.000353 ***
## logwage         9.4104     0.4663  20.180 < 2e-16 ***
## health2. >=Very Good  0.5992     2.5239   0.237 0.812358
## logwage:health2. >=Very Good -0.2760     0.5471  -0.505 0.613928
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 4.61 on 2936 degrees of freedom
## Multiple R-squared: 0.3278, Adjusted R-squared: 0.3272
## F-statistic: 477.4 on 3 and 2936 DF, p-value: < 2.2e-16

##
## Call:
## lm(formula = Resp ~ logwage * maritl, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -26.1661  -1.5995  -0.0348   1.5346  13.9609
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3.7580     1.2111   3.103 0.001934 **
## logwage           5.5804     0.2679  20.828 < 2e-16 ***
## maritlMarried     5.6220     1.4612   3.848 0.000122 ***
## logwage:maritlMarried 0.6256     0.3188   1.962 0.049837 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.672 on 2936 degrees of freedom
## Multiple R-squared: 0.7742, Adjusted R-squared: 0.7739
## F-statistic: 3355 on 3 and 2936 DF, p-value: < 2.2e-16

##
## Call:
## lm(formula = Resp ~ logwage * health_ins, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.4380  -3.4463   0.4241   3.3726  14.1796
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)     -3.6252     1.5696  -2.310 0.021 *
## logwage           8.4247     0.3305  25.494 < 2e-16 ***
## health_ins2. No  -10.0078     2.4300  -4.118 3.92e-05 ***
## logwage:health_ins2. No 2.3026     0.5302   4.343 1.46e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.6 on 2936 degrees of freedom
## Multiple R-squared: 0.3308, Adjusted R-squared: 0.3301
## F-statistic: 483.7 on 3 and 2936 DF, p-value: < 2.2e-16
```

## Appendix 3

```
## Analysis of Variance Table
##
## Model 1: Resp ~ age + maritl
## Model 2: Resp ~ age * maritl
##   Res.Df  RSS Df Sum of Sq    F    Pr(>F)
## 1    2937 26074
## 2    2936 23170  1    2903.9 367.97 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Analysis of Variance Table
##
## Model 1: Resp ~ age + health
```

```
## Model 2: Resp ~ age * health
##   Res.Df   RSS Df Sum of Sq      F   Pr(>F)
## 1    2937 66801
## 2    2936 66612   1    188.85 8.3237 0.003942 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Analysis of Variance Table
##
## Model 1: Resp ~ age + health_ins
## Model 2: Resp ~ age * health_ins
##   Res.Df   RSS Df Sum of Sq      F   Pr(>F)
## 1    2937 67014
## 2    2936 66887   1    126.9 5.5703 0.01833 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#### Appendix 4

```
## Analysis of Variance Table
##
## Model 1: Resp ~ logwage + health_ins
## Model 2: Resp ~ logwage * health_ins
##   Res.Df   RSS Df Sum of Sq      F   Pr(>F)
## 1    2937 62518
## 2    2936 62119   1    398.99 18.858 1.456e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Analysis of Variance Table
##
## Model 1: Resp ~ logwage + maritl
## Model 2: Resp ~ logwage * maritl
##   Res.Df   RSS Df Sum of Sq      F   Pr(>F)
## 1    2937 20988
## 2    2936 20961   1    27.487 3.8501 0.04984 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#### Appendix 5

#### Appendix 6

#### Appendix 7

#### Appendix 8

#### Appendix 9

#### Appendix 10

#### Appendix 11